

Phase D6 Overview and Boundary Case Framework

Derivation Note D6.01 — Series D, Phase D6

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 15, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

This note opens Phase D6 of the Series D derivation program, the final phase before Series CC (cross-paper inheritance edges) and Series T (trilogy-wide ambiguity reduction) begin. Phase D6 addresses boundary cases: governance scenarios at the edges of Paper 3's architectural commitments where the architecture must make explicit choices about what is in scope versus out of scope, where governance requirements at the extreme of their range produce different properties than standard cases, or where a scenario closely resembles Full Aspect Integration (FAI) but is architecturally distinct from it. The note defines boundary cases and their prior-art function, introduces the three boundary case types (minimum, maximum, and boundary definiteness), maps the fourteen subsequent Phase D6 notes to those types, and explains how boundary case precision forecloses the three directions from which scope-based adversarial challenges can come. Phase D6 together with Phase D2 (operational variants, establishing the positive scope) and Phase D3 (anti-patterns, establishing what is architecturally impermissible) constitutes a complete three-part picture of Paper 3's governance domain.

1. Why boundary cases matter for prior art

The derivation notes in Phases D1 through D5 have established Paper 3's architecture from the inside: what the shared substrate is, how Full Aspect Integration operates over it, how the three-tier conflict-handling mechanism works, how the four-locus evolution-feed mechanism routes outputs to home substrates, how configuration-as-substrate-content provides recursive governance coverage, and how population-scale collective evolution constitutes what the preceding claims compose to at population register. That interior work establishes the architecture's positive content.

But prior-art coverage does not conclude with the interior. An adversary challenging the prior art's scope does not need to argue that a core case is different from what the prior art describes — it may instead argue that a particular use case falls just outside the prior art's scope because it is too simple, too complex, or

adjacent-but-distinct. Each of these is a scope-based argument aimed not at the architecture's center but at its edges.

Phase D6 addresses edges directly. Each Phase D6 note takes one boundary scenario and establishes precisely where the architecture's scope boundary runs at that location. The aggregate effect is to render the scope boundary legible from the outside — not as a vague perimeter defined by exemplary cases, but as a specific line whose location is publicly recorded at each point where adversarial challenge is plausible.

The three-phase structure across D2, D3, and D6 provides complete coverage:

Phase D2 established the positive scope — the operational variants and decompositions that are within Paper 3's claimed territory, including the full range of FAI configurations, conflict-handling patterns, evolution-feed routings, and governance dimensions.

Phase D3 established the impermissible — the anti-patterns that violate Paper 3's architectural commitments even if they superficially resemble compliant configurations.

Phase D6 establishes the scope boundary — the exact location of the line between what is and is not a Paper 3 FAI event, what is and is not a compliant shared substrate, and what adjacent scenarios are architecturally distinct from the claimed architecture.

Together, these three phases answer the three questions a complete prior-art record must answer: what is within scope, what is impermissible within scope, and where the scope ends.

2. What boundary cases are

A boundary case is a governance scenario at the edge of Paper 3's architectural commitments that satisfies one or more of the following conditions:

Condition A — In-scope determination required. The scenario sits close to the scope boundary, and the architecture must make explicit whether the scenario is within or outside FAI's claimed territory. These cases are not obviously compliant and not obviously out-of-scope; the boundary case note records the explicit determination.

Condition B — Range-extreme properties. The scenario involves a governance requirement at its minimum or maximum, where behavior at the extreme differs structurally from behavior in the standard mid-range case. The scenario is within scope, but its extreme-case properties need explicit articulation so that the extreme is not mistaken for a separate architecture.

Condition C — Adjacent-but-distinct scenario. The scenario closely resembles FAI in surface form — it involves coordination among distinct organizational entities, or content exchange between structured substrates, or some other feature associated with Paper 3 — but falls outside FAI's claimed territory on one or more architectural dimensions. These cases establish the negative boundary: the specific reasons a scenario that resembles FAI is nonetheless not FAI.

Not every scenario that touches Paper 3's architecture is a boundary case. Standard operational cases — the representative mid-range configurations documented in Phase D2 — are not boundary cases. Anti-patterns — configurations that violate Paper 3's commitments in ways that are clear from the commitment's content — are not boundary cases. Boundary cases are specifically the scenarios where the scope line needs to be explicitly drawn because proximity to the line makes it otherwise ambiguous.

3. Three boundary case types and their adversarial-foreclosure functions

Type A — Minimum boundary cases

Minimum boundary cases establish the floor of Paper 3's scope: the simplest thing that qualifies as a compliant FAI event, the minimal properties a shared substrate must exhibit, and the simplest compliant participant. These cases answer: how little is enough for the architecture to apply?

The adversarial challenge they foreclose is the "too simple to be within scope" argument. An adversary might claim that a minimally-featured coordination event — one aspect contributed, two Selves, one governance decision, no conflict — falls outside the prior art's claimed territory because it lacks the complexity the prior art's representative examples exhibit. Minimum boundary cases foreclose this by recording the explicit floor: what properties are necessary and sufficient for the architecture to apply, with no requirement of further complexity.

Three notes cover minimum boundary cases: D6.02 (minimum valid FAI event), D6.03 (minimum valid shared substrate), and D6.04 (minimum valid participation).

Type B — Maximum boundary cases

Maximum boundary cases establish the ceiling of what the architecture's governance mechanisms can handle: the most comprehensive sharing scope, the highest cardinality, the most extensive governance record. These cases answer: how much complexity can the architecture govern?

The adversarial challenge they foreclose is the "too complex / too large to be within scope" argument. An adversary might claim that a large-scale, high-cardinality, fully-recursive governance configuration exceeds what the prior art's architecture can accommodate — that the prior art describes only modest configurations and the extreme is a new architecture not covered by the prior art. Maximum boundary cases foreclose this by recording the architecture's depth ceiling and establishing that the ceiling is a property of the same architectural commitments, not a departure from them.

Three notes cover maximum boundary cases: D6.05 (maximum sharing scope — full-Self contribution), D6.06 (maximum cardinality — how many Selves can participate in one FAI event), and D6.07 (maximum governance depth — the most extensive compliant governance record).

Type C — Boundary definiteness cases

Boundary definiteness cases address scenarios that are adjacent to FAI but fall outside its scope, or that sit at the edge of FAI's claimed territory. These cases answer: what makes this scenario architecturally distinct from FAI, and where precisely is the line?

The adversarial challenge they foreclose is the "adjacent but different" argument — a challenge that runs in the opposite direction from the minimum and maximum challenges. Here the adversary does not argue that the claimed architecture fails to cover a scenario; instead, the adversary argues that a scenario the prior art appears to cover is actually a different architecture that the prior art does not reach. Boundary definiteness cases foreclose this by establishing the exact architectural criteria that distinguish FAI from adjacent but distinct scenarios.

Eight notes cover boundary definiteness cases: D6.08 through D6.14, addressing the ungoverned vs. governed coordination boundary, the intra-Self vs. inter-Self boundary, the aspect vs. Self contribution boundary, the temporary vs. persistent shared substrate boundary, the two-Selves vs. population scope transition, the cooperation vs. competition boundary, and the FAI vs. merger/acquisition governance distinction.

4. The complete Phase D6 note sequence

The fourteen boundary case notes proceed in type order: minimum cases first, maximum cases second, boundary definiteness cases third. Within each type, the notes proceed from the most foundational boundary to the most contextual.

Type A — Minimum boundary cases (D6.02–D6.04)

D6.02 — Minimum valid FAI event: What is the simplest configuration that constitutes a compliant FAI event? Establishes the floor below which a coordination scenario is not within Paper 3's scope.

D6.03 — Minimum valid shared substrate: What properties must a shared substrate exhibit to qualify as a shared substrate under Paper 3's Claim 1? Establishes the property floor below which a coordination medium is not a shared substrate in the Paper 3 sense.

D6.04 — Minimum valid participation: What does the simplest compliant participant — a Self contributing to an FAI event — look like? Establishes the participation floor below which a participant is not a compliant FAI contributor.

Type B — Maximum boundary cases (D6.05–D6.07)

D6.05 — Maximum sharing scope: What is the most comprehensive aspect contribution a Self can make within one FAI event, up to full-Self contribution? Establishes the contribution ceiling and the governance requirements it triggers.

D6.06 — Maximum cardinality: How many Selves can participate in a single FAI event, and what architectural properties hold at maximum cardinality? Establishes the cardinality ceiling and its governance implications.

D6.07 — Maximum governance depth: What is the most extensive compliant governance record a shared substrate can carry, from minimal persistence to full-substrate retention with complete provenance? Establishes the depth ceiling and the recursive governance properties it activates.

Type C — Boundary definiteness cases (D6.08–D6.14)

D6.08 — Ungoverned vs. governed coordination: The exact architectural line between FAI (governed inter-Self coordination over a shared substrate under joint human authority) and ungoverned inter-organizational coordination (the named foil established in D1.05). Forecloses arguments that ungoverned coordination is within scope, and symmetrically forecloses arguments that FAI is merely a labeled version of generic inter-organizational coordination.

D6.09 — Intra-Self vs. inter-Self boundary: When does coordination between aspects within one Self cross into inter-Self FAI territory? Establishes the perimeter criterion that distinguishes Paper 2's intra-Self scope from Paper 3's inter-Self scope, so that the two scopes are precisely delimited at their shared boundary.

D6.10 — Aspect vs. Self boundary in contribution: When does an aspect's contribution to a shared substrate constitute a Self-level contribution, and when does it remain at aspect scope? Establishes the contribution-scope criterion that governs which governance requirements apply.

D6.11 — Temporary vs. persistent shared substrate: The shared substrate is temporarily constructed for an FAI event by architectural default, but persistence is governance-configured per event. This note establishes when persistence becomes architecturally significant and what governance requirements it triggers at each persistence level.

D6.12 — Two Selves vs. population scope: Bilateral FAI (the minimum cardinality case) and population-scope collective evolution (Paper 3's Claim 6) are both within scope, but they are not the same architectural phenomenon. This note establishes the transition criteria — when accumulated bilateral FAI becomes population-scope coordination and what governance architecture that transition activates.

D6.13 — Cooperation vs. competition boundary: FAI is a cooperation architecture, but Selves that cooperate through FAI events may also compete in the same domain. This note establishes where the architectural line between a cooperation event and a competition event sits, and what governance applies at the boundary.

D6.14 — FAI vs. merger/acquisition governance: FAI involves content exchange between distinct Selves under joint authority, and organizational merger/acquisition involves structural integration of organizations under a new governance perimeter. The surface resemblance masks a deep architectural distinction. This note establishes the architectural criteria that distinguish FAI governance from merger/acquisition governance, foreclosing both conflation and scope exclusion.

Phase D6 closure (D6.15)

D6.15 — Phase D6 final closure and Series D synthesis: Closes Phase D6 and synthesizes the complete Series D derivation program, establishing what the full set of Series D notes — Phases D0 through D6, covering claim anchors, foundational sub-commitments, operational variants, anti-patterns, composition pairs, operational tests, and boundary cases — collectively establish as Paper 3's prior-art record.

5. Phase D6 as the final Series D phase

Phase D6 occupies a specific position in the derivation program's architecture. Series D runs from Phase D0 (Paper 3 claim anchors) through Phase D6 (boundary cases). Each phase adds a layer of prior-art coverage:

Phase D0 anchors the claims.

Phase D1 establishes the foundational sub-commitments.

Phase D2 maps the operational variants — the full range of compliant configurations within scope.

Phase D3 maps the anti-patterns — configurations that violate the commitments while resembling compliant ones.

Phase D4 addresses composition pairs — how Paper 3 commitments compose with each other and with Paper 1 and Paper 2 commitments.

Phase D5 provides operational tests — the discriminating criteria by which a given system is evaluated against the architecture's commitments.

Phase D6 closes the boundary — the explicit scope-edge record at every point where adversarial challenge is plausible.

After D6.15, Series D is complete. The subsequent series — Series CC (cross-paper inheritance edges, formalizing what Paper 3 inherits from Papers 1 and 2 without redefense) and Series T (trilogy-wide ambiguity reduction, disambiguating reused terms across the three papers) — operate at the cross-paper level. They require Series D to be complete because they build on the single-paper prior-art record Series D establishes.

Phase D6's position as the final Series D phase means that the boundary case notes carry a closure responsibility beyond their individual scope: together, they constitute the final layer of Paper 3's standalone prior-art coverage. The scope boundary Phase D6 draws is the boundary that Series CC and Series T will subsequently treat as settled when they extend the derivation work across papers.

6. Operational test

A derivation note belongs in Phase D6, rather than in Phases D2, D3, D4, or D5, if and only if all of the following are true:

1. The note addresses a scenario at or near the scope boundary of Paper 3's architectural commitments — a scenario where reasonable prior-art analysis might place the scenario either within or outside scope without the explicit boundary determination the note provides.
2. The note's primary contribution is the explicit determination of where the scope boundary runs at that scenario's location, not a positive description of a compliant configuration (Phase D2), a prohibition of an impermissible configuration (Phase D3), a composition relationship (Phase D4), or a discriminating test criterion (Phase D5).
3. The note forecloses at least one of the three scope-based adversarial challenge types: too-simple (Type A), too-complex (Type B), or adjacent-but-distinct (Type C).

A note that fails any of (1)–(3) belongs in a different phase, or is not a boundary case note at all.

7. Conclusion

Phase D6 completes Series D by establishing Paper 3's scope boundary at each point where adversarial challenge is plausible. The fourteen boundary case notes address three challenge directions — minimum (floor-based), maximum (ceiling-based), and boundary definiteness (lateral) — across the three-tier typology of Conditions A, B, and C. Together with Phase D2's positive scope record and Phase D3's anti-pattern record, Phase D6 constitutes the final layer of Paper 3's standalone prior-art coverage.

Subsequent work in Series CC and Series T will treat the scope boundary Phase D6 establishes as settled, building cross-paper coverage on the foundation Phase D6 closes.

Source paper

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *Phase D6 Overview and Boundary Case Framework*. Derivation Note D6.01. May 15, 2026.
ORCID: 0009-0004-8065-3235.